

Houssam Hamrouni

Software Engineer specializing in AI & Security

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PROFESSIONAL SUMMARY

Product-Minded Software Engineer with an MSc in Cybersecurity and 1.5+ years of experience engineering secure, scalable systems. Specializes in AI Agents (Multimodal/Voice), IoT, and Web3 architectures. Proven ability to bridge the gap between theoretical research and production-grade software, delivering complex solutions like real-time digital twins, anti-fraud ticketing systems, and automated medical receptionists.

TECHNICAL SKILLS

- Languages: TypeScript, Python, C#, Rust, Solidity, SQL.
- Frontend: Next.js 15 (App Router), React 19, Tailwind CSS, React Native (Expo).
- Backend: Node.js, ASP.NET Core, Django, Supabase (PostgreSQL + RLS), Redis.
- AI & Data: TensorFlow, OpenAI Realtime API, Gemini Live API, Computer Vision (OpenCV).
- Security & Web3: OAuth2/OIDC, Argon2id Encryption, ERC-4337 (Account Abstraction), Slither.
- Infrastructure: Docker, CI/CD, Vercel Edge Functions, Raspberry Pi (IoT).

PROFESSIONAL EXPERIENCE

Software Engineer	Jan 2024 – Present
Frederick University (Remote / Hybrid)	Nicosia, Cyprus

- Project: FoodFlow (IoT Smart Waste System)
 - Built a production-grade Cyber-Physical System for commercial kitchens to track food waste, competing with market leaders.
 - Developed edge computing software for Raspberry Pi using Python & PyQt5, integrating Gemini Vision API to automatically classify food types and weight in real-time.
 - Engineered a multi-tenant SaaS dashboard using Django and PostgreSQL, providing real-time analytics that reduced kitchen waste by ~15% in pilot tests.
- Project: INVESTech (Ed-Tech Enterprise Platform)
 - Architected the backend for a pan-European vocational excellence platform using ASP.NET Core and SQL Server.
 - Implemented enterprise security features including Role-Based Access Control (RBAC) and secure file handling for sensitive EU compliance data.
 - Engineered a localized content management system supporting 10+ languages.

FEATURED ENGINEERING PROJECTS

ClinicVoice – AI Voice Receptionist | TypeScript, Gemini Live, Supabase

- Engineered a real-time B2B voice agent for medical clinics capable of handling inbound calls, answering FAQs, and booking appointments with <500ms latency.
- Implemented full-duplex audio streaming using WebSockets and Gemini Live API for "barge-in" capabilities.
- Built a robust tool-calling system for real-time database writing and calendar querying.
- Commercial Viability: Designed to recapture 30% of missed clinic revenue with operating costs of <€0.10 per call.

TicketChain – "Invisible" Web3 Ticketing | Next.js 15, Solidity, Base L2

- Built a SaaS ticketing platform solving crypto UX friction via Account Abstraction (ERC-4337) and Paymasters, allowing gasless NFT ticket purchases via credit card.
- Developed an offline-first scanner app (React Native) using rotating HMAC-SHA256 QR codes (TOTP style) to prevent screenshotting and replay attacks.
- Implemented a "Gate Lock" smart contract modifier to freeze transfers 2 hours before events.

Secure Password Manager | Rust, Electron, Argon2id, SQLite

- Developed a local-first security tool focusing on memory safety.
- Utilized Rust FFI and the **secrecy** crate to ensure zeroization of memory buffers, preventing sensitive data leaks to swap files or core dumps.

EDUCATION

Master in Cybersecurity Exp. 2026
European University Cyprus Focus: Application Security, Cryptography, Secure Architecture.

Bachelor in IT & Computer Science 2021 – 2024
La Rochelle Université France

AWARDS & CERTIFICATIONS

- 2024: Participant: Cyprus Tech Odyssey (XRPL Hackathon)
- 2023: 2nd Place Winner: British Council AI Hackathon ("AI for the Common Good")
- Cert: Cisco Cybersecurity Essentials